类设计

1. **Person类**
    - 属性：唯一标识 id
    - 功能：基类功能
2. **Employee类**（继承自Person）
    - 额外属性：salary（月薪）
    - 工资计算：Employee_Salary = salary
3. **Manager类**（继承自Employee）
    - 额外属性：baseSalary（基本工资）和 bonusRate（奖金系数）
    - 工资计算：Manager_Salary = baseSalary + (baseSalary * bonusRate)
4. **Company类**
    - 存储员工信息
    - 初始化参数：number（公司人数）
    - 功能：
        - operator[] 访问员工
        - saveToFile() 保存信息
        - 支持多态调用printInfo()输出信息
        - 能释放新个人文件

要求

- 实现多态调用printInfo()输出信息
- 实现工资计算功能
- 实现文件保存功能
- 需要编写main函数进行调试

参考答案

```
#include <iostream>
#include<ctime>
#include<cstdlib>
#include <fstream>

using namespace std;

class Person{
public:
    int id;
    Person(int i) : id(i) {}
    virtual void printObj() = 0;
};

class Employee: public Person{
```

```
public:
    double salary;
    Employee(int i, double s): Person(i),salary(s) {}
    void printObj(){
        cout << " Employee:" <<id<< " Salary:" <<salary<<endl;
        return ;
    }
};

class Manager: public Person{
public:
    double basicSalary,bonusRate;
    double salary;
    Manager(int i, int bs, int br) : Person(i),basicSalary(bs),bonusRate(br) {
        salary = basicSalary + bonusRate*basicSalary;
    }
    void printObj(){
        cout << "Manager:" << id << " Salary:" << salary<<endl;
        return ;
    }
};

class Company{
public:
    Person ** person;
    int size;
    Company(int n){
        person = new Person* [n];
        size = n;
    }
    ~Company(){
        delete []person;
    }
    Person *& operator[](int index){
        return person[index];
    }
};
int main() {
    Company company(4);
    company.person[0]= new Employee(1,100);
    company.person[1] = new Employee(2,200);
    company.person[2] = new Manager(3, 100, 200);
    company.person[3] = new Manager(4,100,300);
    for(int i = 0; i < 4; i++){
        company.person[i]->printObj();
    }
}
```